

Subjective Expected Utility Representation

Primitives:

$\mathbb{X}$ a nonempty set of outcomes (not necessarily finite).

$\mathcal{S} = \{1, \dots, S\}$, a set of states.

$\mathbb{X}^{\mathcal{S}}$, a set of *acts*.

$\succsim \subset \mathbb{X}^{\mathcal{S}} \times \mathbb{X}^{\mathcal{S}}$, a binary relation on $\mathbb{X}^{\mathcal{S}}$ (with $\succ$ and $\sim$ defined in the usual way).

If $\mathbb{X}$ is a connected subset of a Euclidean space and $\succsim$ is complete, transitive, and continuous, then there is a continuous representation U of $\succsim$ (Debreu, 1959, *Theory of Value*, Section 4.6).

In the opening passage of Chapter 7 of Debreu's *Theory of Value*, he writes

The analysis is extended... to the case where uncertain events determine the consumption sets, the production sets, and the resources of the economy. A contract for the transfer of a commodity now specifies, in addition to its physical properties, its location and its date, an *event* on the occurrence of which the transfer is conditional. This *new* definition of a commodity allows one to obtain a theory of uncertainty *free from any probability concept* and formally identical with the theory of certainty developed in the preceding chapters [Debreu, 1959, p. 98, emphases added].

In particular I note that the First Welfare Theorem holds in this setting. But this version requires there to be markets for each good conditional on each event.

Debreu strongly suggests we should use this general representation—which does not imply expected utility or even *probabilistic sophistication* (that a consumer's choices are consistent with some belief about the states). By contrast, almost the entire profession uses a subjective expected utility representation in applications. So it merits a definition.

Definition (Subjective Expected Utility). *A representation U of $\succsim$ is a **subjective expected utility representation** of $\succsim$ if there is a real-valued function u on $\mathbb{X}$ and a probability distribution $\pi(\cdot)$ on $\mathcal{S}$ such that*

$$U(x) = \sum_{s=1}^S u(x_s) \pi(s)$$

where $x = (x_1, \dots, x_S)$.

As mentioned, such a representation is not consistent with the “modal” choices in the Ellsberg paradox.